Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name:Rashi Bhatia				**Subject: ASD&D**
TE - 1	**Batch: T11**	**Sem: 5**	**Roll No:12**	**Date:4/8/2026**

EXPERIMENT 04

Aim : To understand Continuous Integration, Jenkins installation, and how to build a declarative pipeline using Jenkins.

LO Mapped : 03

LO : To setup a continuous Integration Pipeline using Jenkins.

Theory:

1. Introduction to Continuous Integration (CI)

Continuous Integration is a cornerstone practice in modern DevOps Methodology. [1]

- The Core Mechanism: In traditional workflows, developers integrated their source code late in the software development lifecycle (SDLC), resulting in major merge conflicts. CI replaces this by requiring developers to commit small code changes to a shared repository daily or hourly. [1]
- Automated Feedback Loop: Every single code push automatically triggers an independent build process alongside unit and integration tests. Bugs and structural breaks are isolated within minutes of being written, drastically lowering debugging costs. [1, 2]

2. Jenkins Architecture & Ecosystem

Jenkins functions as a self-contained, open-source automation server built to orchestrate end-to-end CI/CD workflows. [1]

- Extensibility: It relies heavily on an active community and an ecosystem of over 1,800 modules and plugins. These plugins bridge the server with tools like version control systems (Git), compile systems (Maven, Gradle), container environments (Docker), and cloud providers. [1]
- Distributed System Execution: Jenkins operates on a master-agent architecture. The core instance manages user configurations, scheduling, and UI processing, while offloading resource-heavy build and test executions to remote execution systems or agents. [1, 2]

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3. Understanding the Jenkins Pipeline Blueprint

A Jenkins Pipeline acts as a visual and programmable representation of your continuous deployment model. It views your software movement through sequential, repeatable automation scripts. [1, 2]

[Source Commit] ⟶ 🚂 [Build Stage] ⟶ 🛠️ [Test Stage] ⟶ 🚀 [Deploy Stage]

The underlying concept introduces Pipeline as Code, which allows teams to commit their build infrastructure logic directly inside a configuration file (traditionally named a Jenkinsfile) right beside their application source code. This makes the deployment process transparent, version-controlled, and instantly reproducible. [1, 2]

4. Core Pipeline Concepts and Structural Blocks

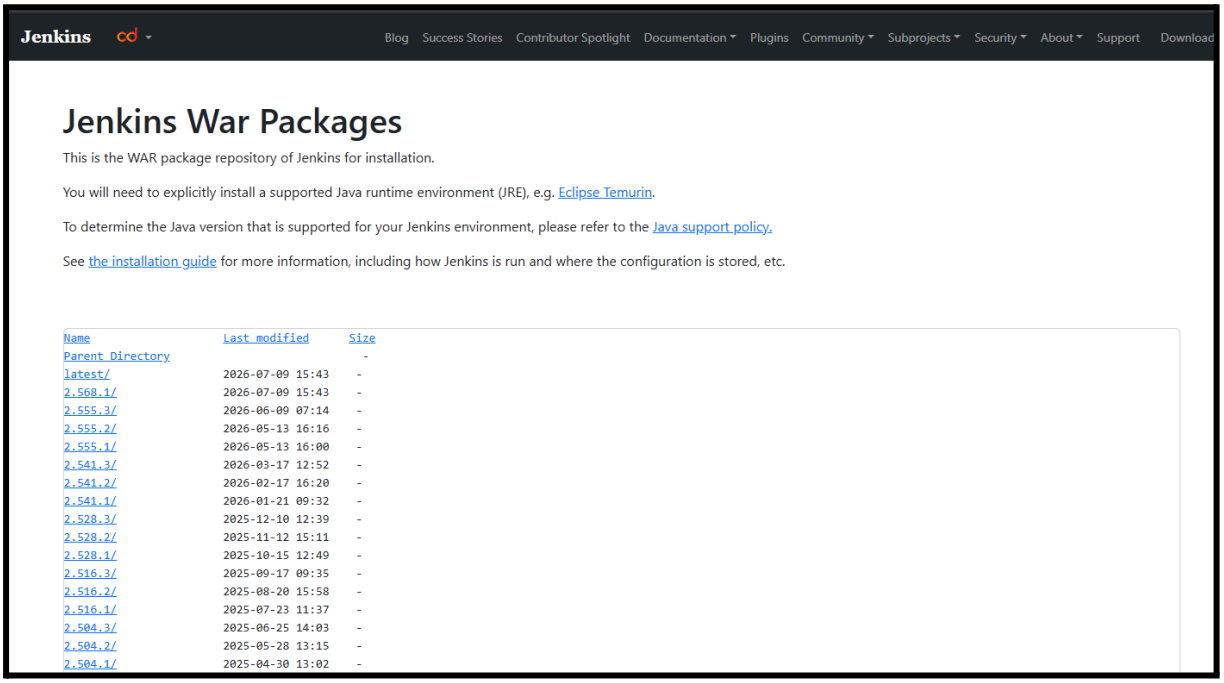
A standardized pipeline contains four fundamental structural building blocks: [1]

- Pipeline / Node: The primary block encompassing the entirety of the build cycle logic.
- Agent: Specifies the exact machine executor where the server will run the runtime tasks. Using agent any tells Jenkins to clear the job for any available running worker environment.
- Stage: A distinct visual segment dividing your workflow phases sequentially (e.g., stage('Build'), stage('Test'), stage('Deploy')).
- Step: An atomic execution unit nested cleanly inside a stage telling the runner a single exact operational task to perform (e.g., executing a command or printing text via echo).[1, 2, 3]

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Installation:



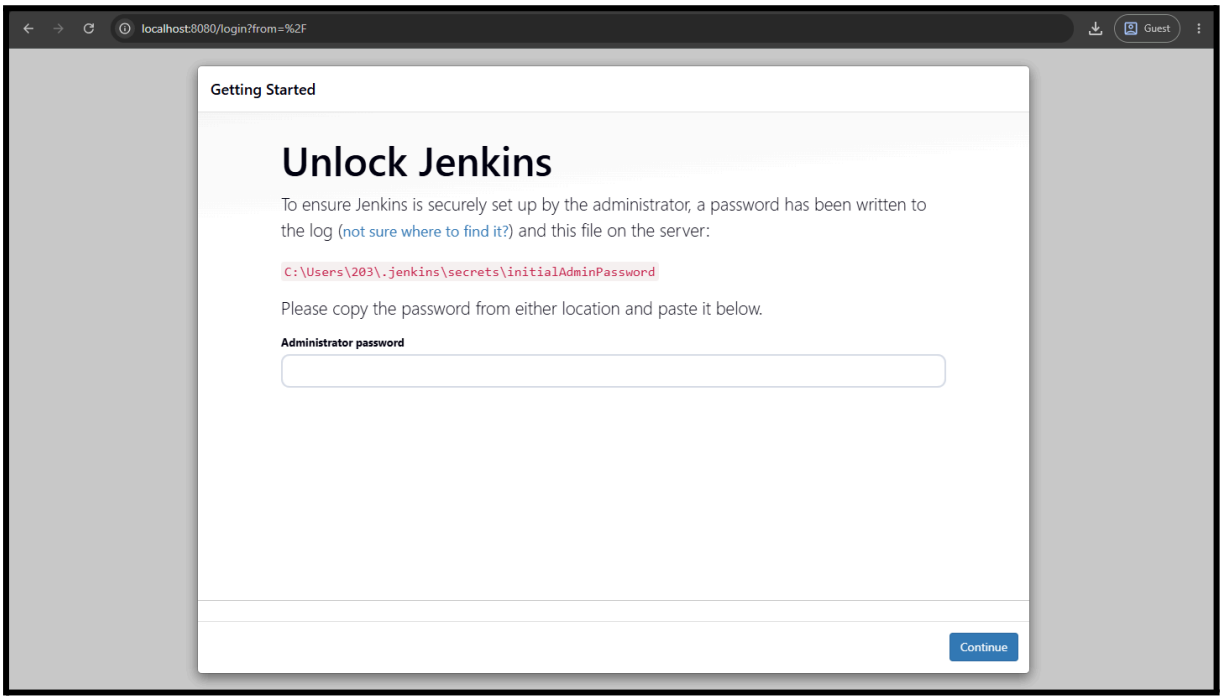


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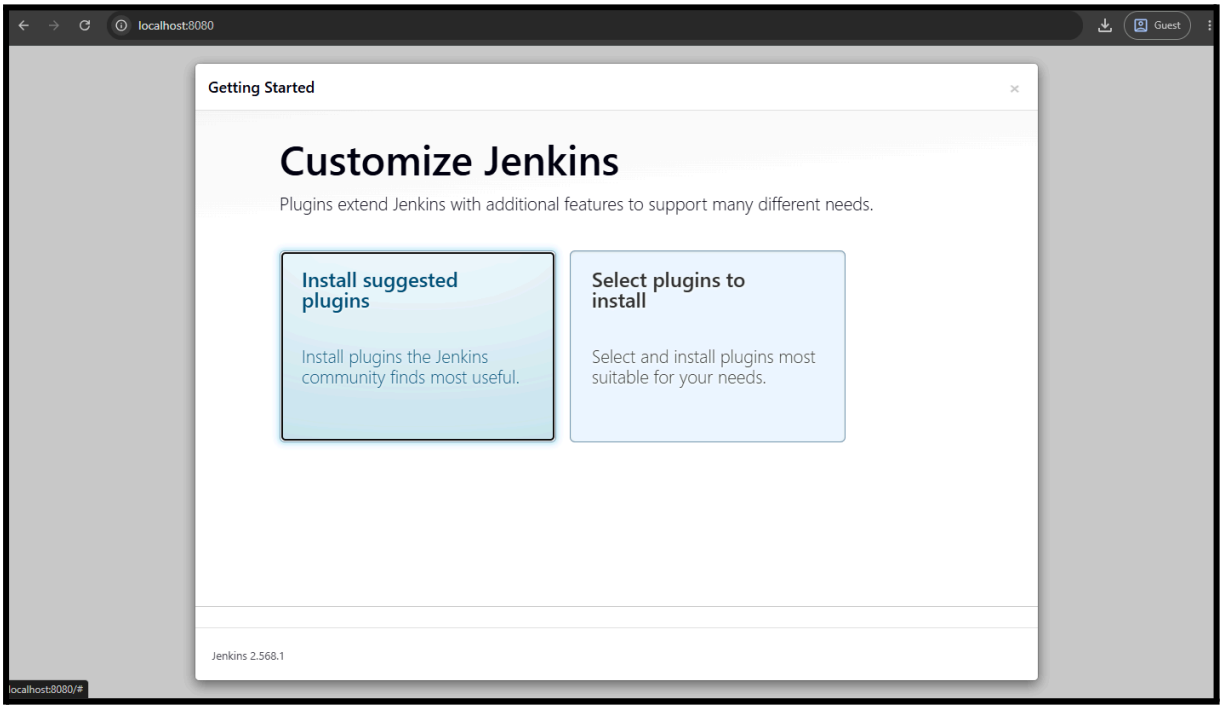
```
C:\Users\203>cd documents

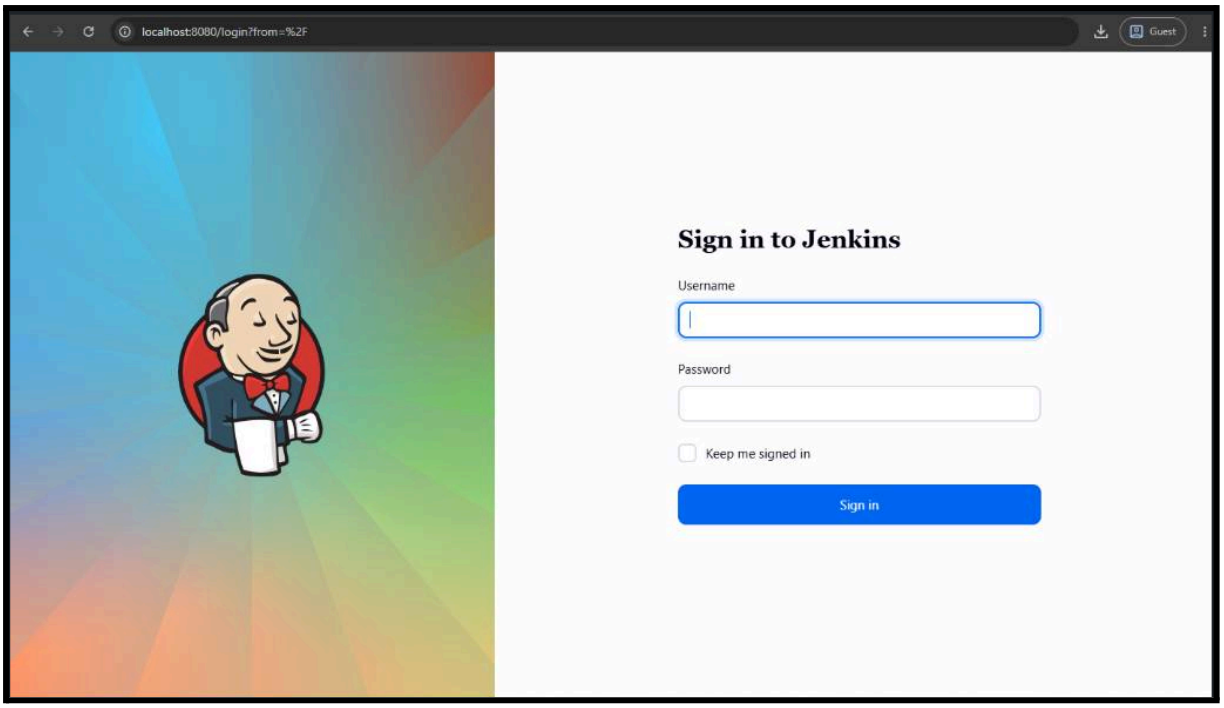
C:\Users\203\Documents>java -jar jenkins.war
```



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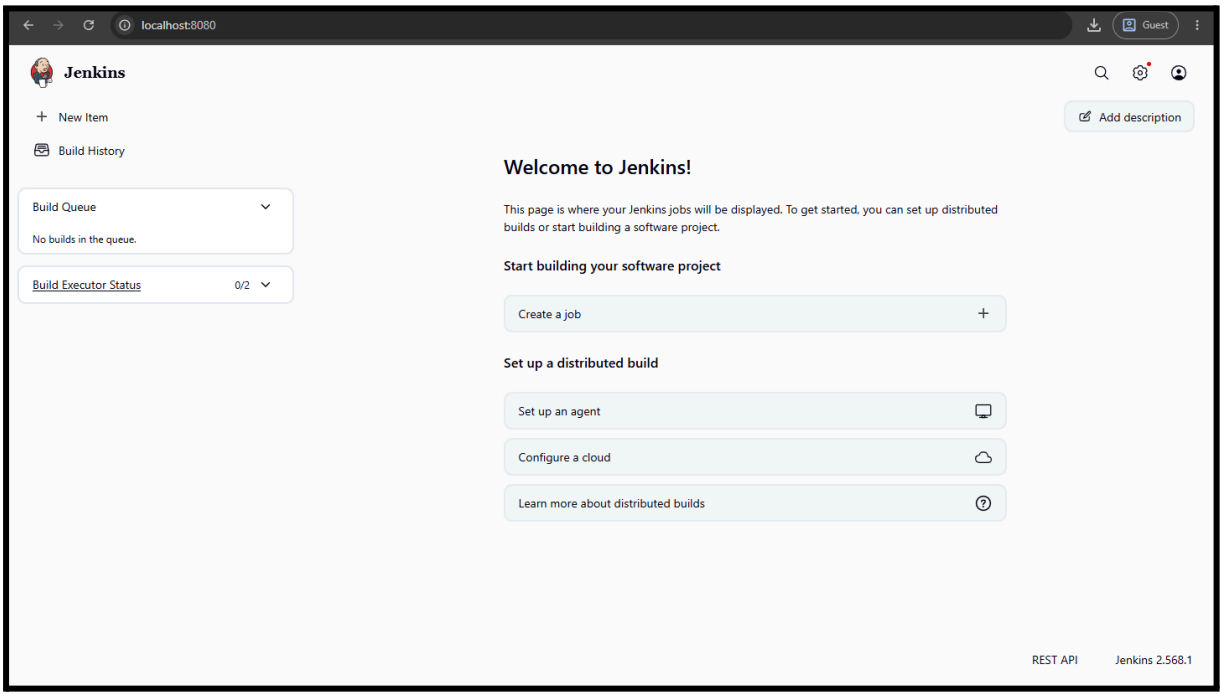
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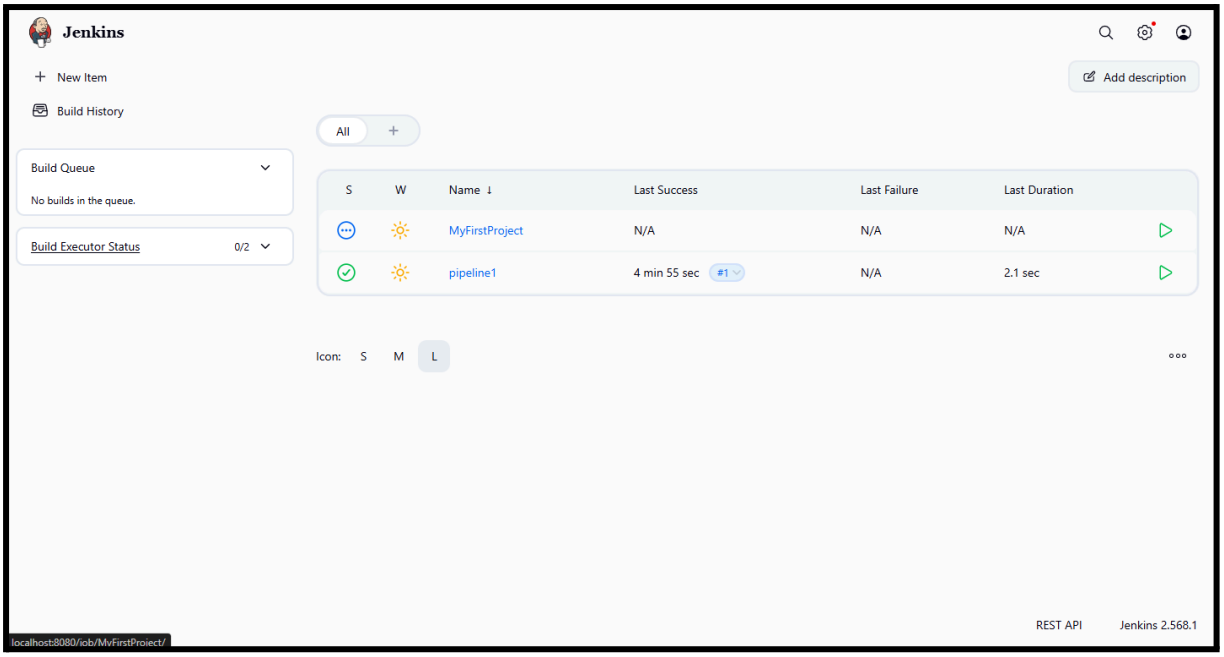




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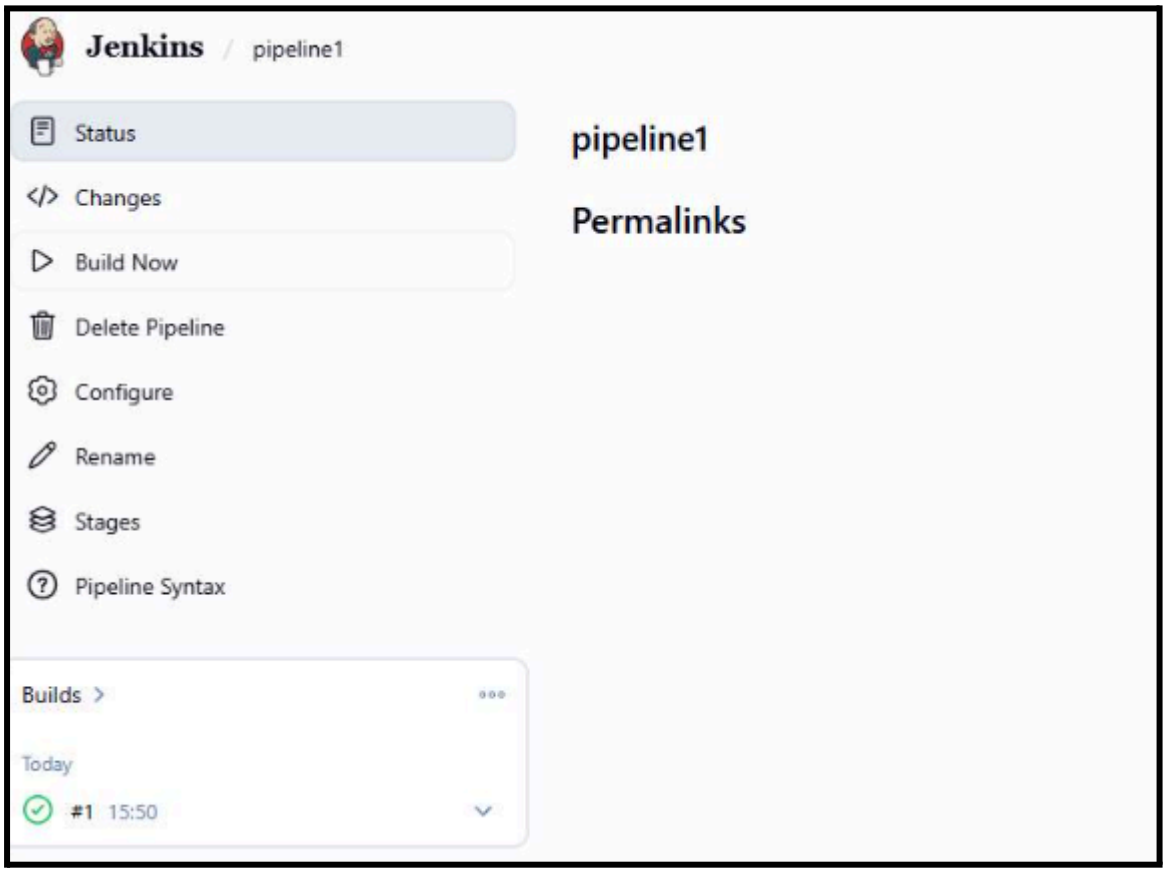
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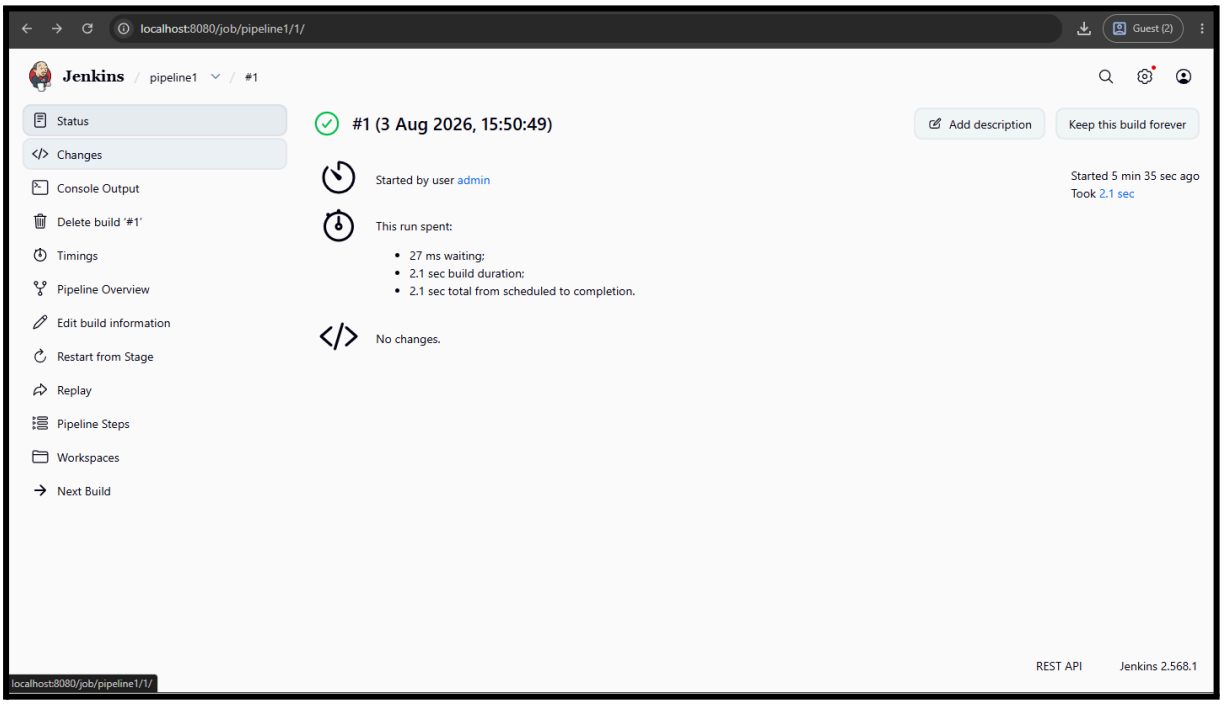
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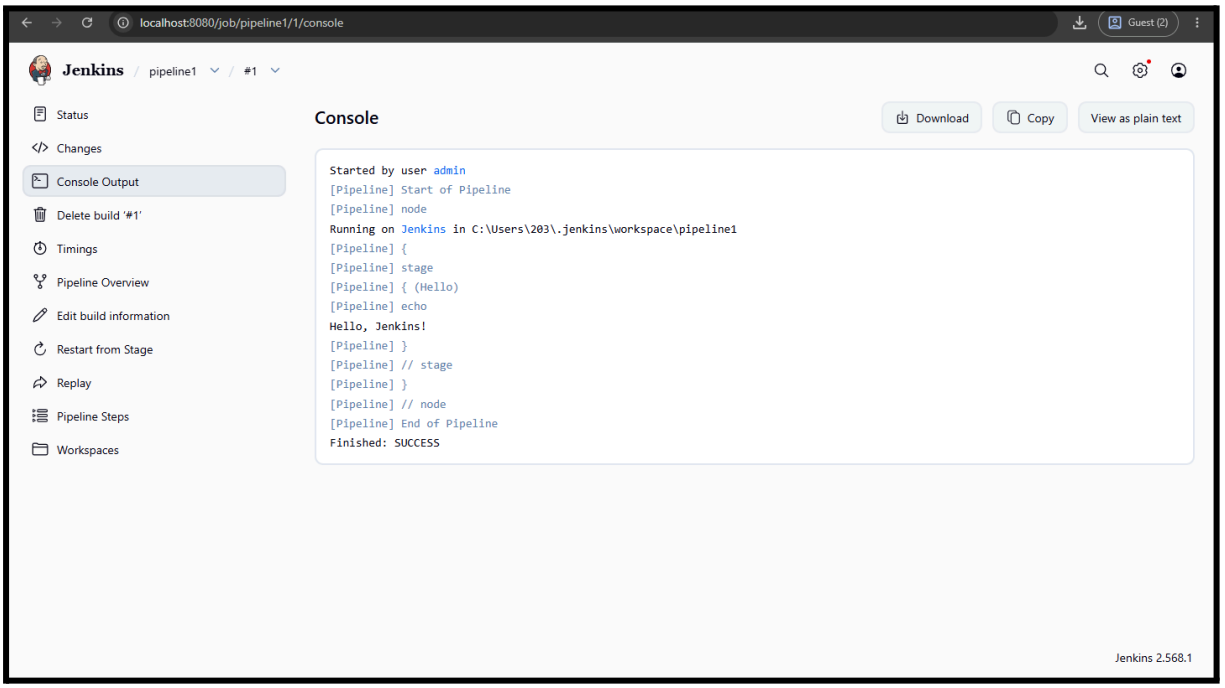
Code:

```
pipeline {
    agent any
        stages {
            stage('Hello') {
                steps {
                echo "Hello, Jenkins!"
                }
        }
    }
}
```

Output:

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Conclusion

In this assignment, we successfully learned about Continuous Integration and the step-by-step method to install and unlock Jenkins. We explored both Declarative and Scripted pipelines to understand their structure and benefits. By implementing a basic Declarative Pipeline, we demonstrated how Jenkins executes automated build steps to enable efficient, repeatable, and reliable CI/CD workflows.